

E-Commerce Sales Performance Analysis

Using Amazon Sales Data

AICTE | IBM SkillsBuild Data Analytics with AI Internship Program 2026

1. Executive Summary

This project analyzes 128,975 records covering 31 March 2022 to 29 June 2022. Recorded sales total approximately **₹7.86 crore** and the average recorded order amount is **₹648.56**. The calculated cancellation rate is 14.21%. The analysis covers product categories, monthly trends, states, fulfilment, order status and B2B/B2C activity.

2. Problem Statement

The project converts raw e-commerce transaction records into useful information for understanding sales performance, product demand, geographic contribution and operational signals.

3. Objectives

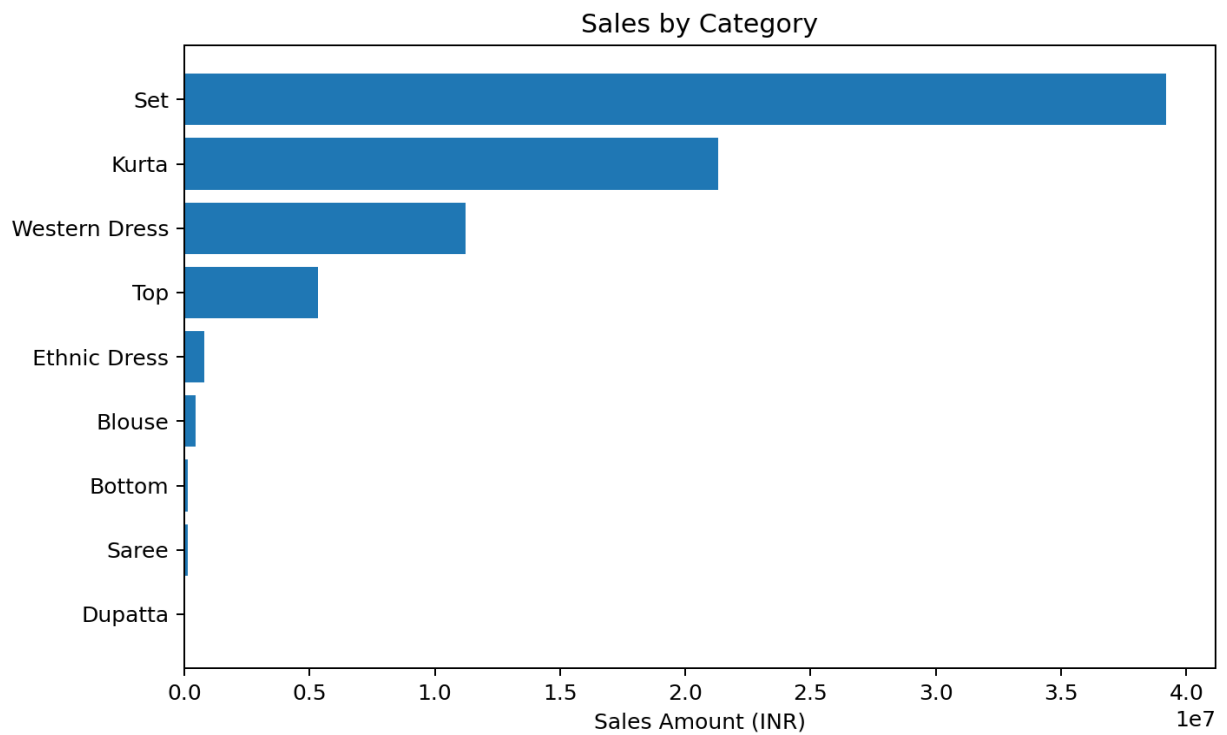
- Analyze sales and average order amount.
- Identify high-performing categories and SKUs.
- Understand monthly and geographic sales patterns.
- Compare fulfilment and B2B/B2C activity.
- Identify cancellation-related signals and formulate recommendations.

4. Key KPIs

KPI	Result
Records	128,975
Total recorded sales	₹78,592,678.30
Average order amount	₹648.56
Cancellation rate	14.21%
Amazon fulfilment sales	₹54,322,151.00
Merchant fulfilment sales	₹24,270,527.30

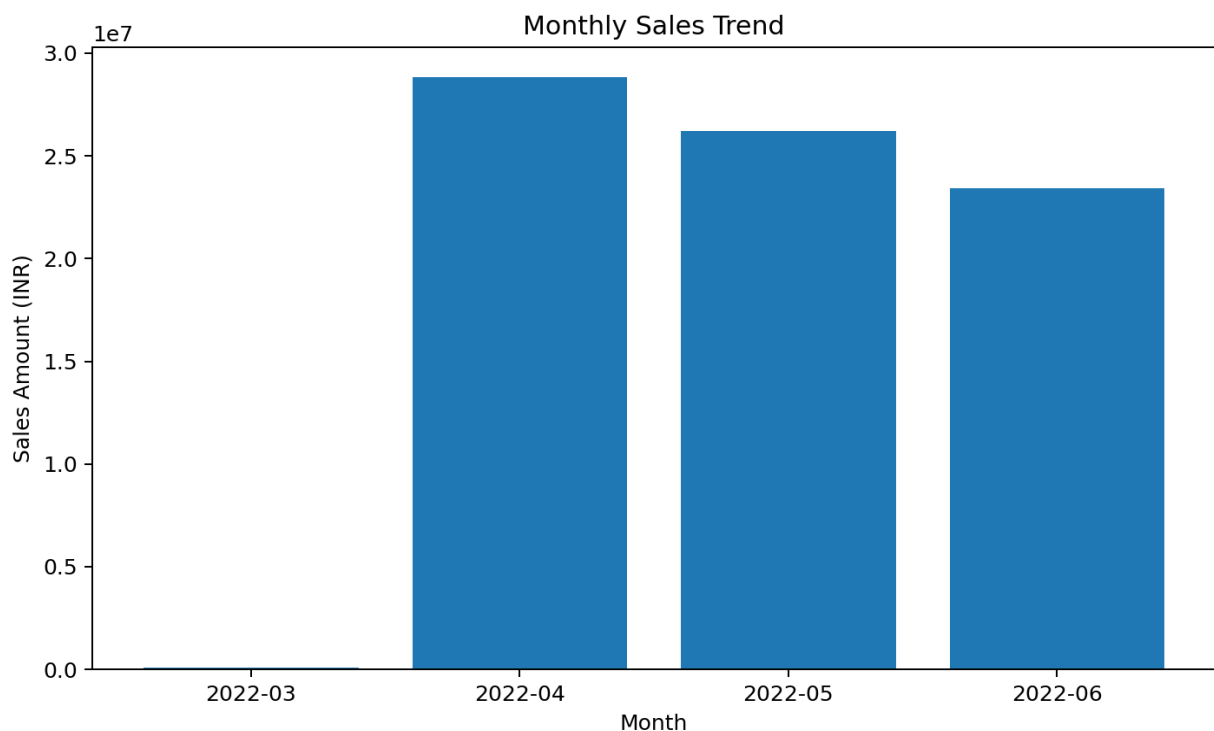
5. Category Sales

Set and Kurta are the largest recorded sales categories, followed by Western Dress and Top.



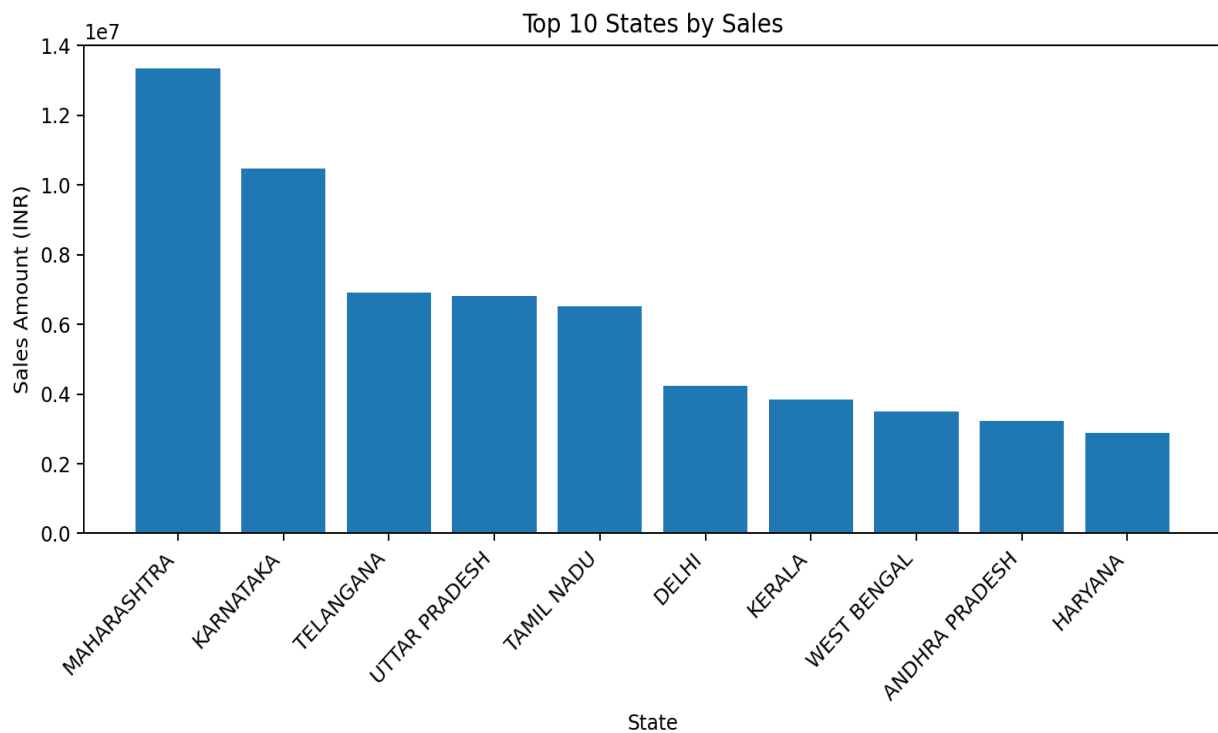
6. Monthly Sales Trend

Recorded sales increase sharply from March to April and then decline in May and June. March represents only a short portion of the dataset period.



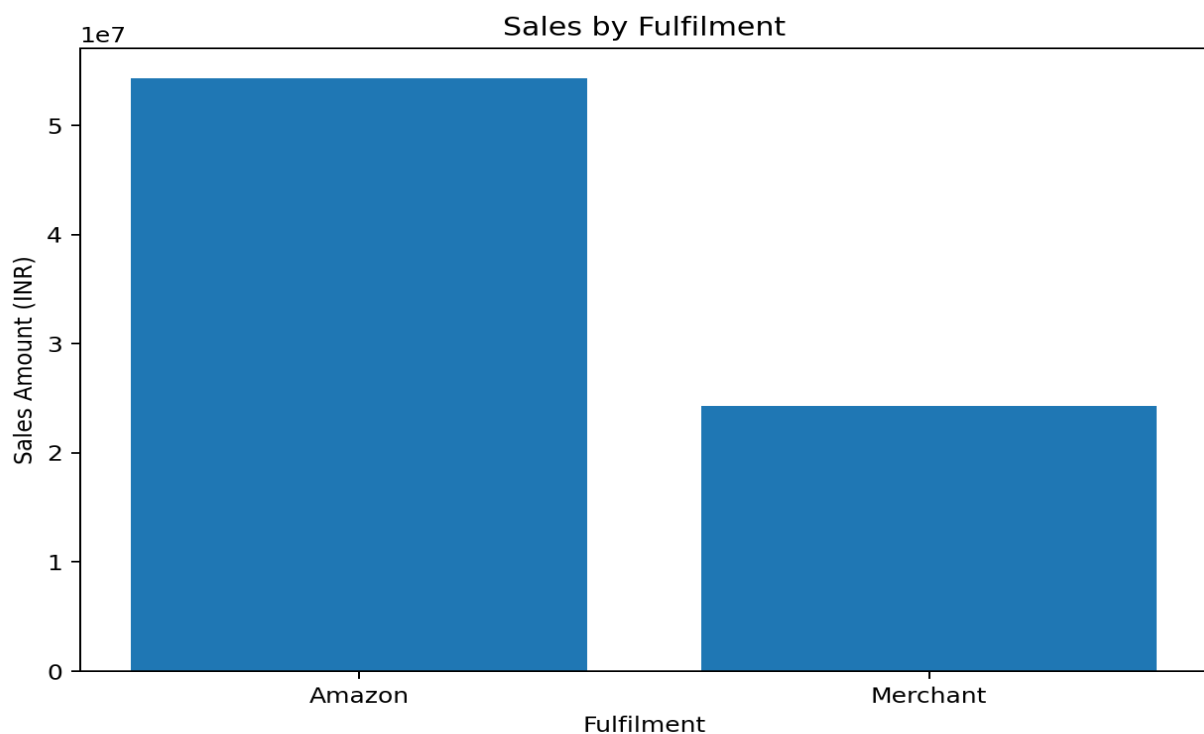
7. Geographic Sales

Maharashtra has the highest recorded sales among the top ten states, followed by Karnataka.



8. Fulfilment Analysis

Amazon fulfilment has a higher recorded sales amount than Merchant fulfilment in this dataset.



9. Key Insights and Recommendations

- Sales are concentrated in a small number of product categories, particularly Set and Kurta.
- The dataset shows a strong increase in recorded sales from March to April, followed by lower totals in May and June.
- Maharashtra and Karnataka are the two highest-sales states among the top ten states analyzed.
- Amazon fulfilment accounts for a larger recorded sales amount than Merchant fulfilment.
- The recorded cancellation rate is 14.21%, making cancellation monitoring relevant for operational analysis.
- B2C transactions account for substantially more recorded sales than B2B transactions in this dataset.

- Monitor Set and Kurta inventory and availability because these categories contribute large recorded sales amounts.
- Investigate cancellation reasons by product, state, fulfilment method and time period to identify operational patterns.
- Review performance in high-sales states and compare logistics, inventory availability and customer demand across regions.
- Track monthly sales using a consistent time window so that partial months are not compared directly with full months.
- Monitor Amazon and Merchant fulfilment separately using delivery, cancellation and return metrics rather than sales alone.
- Use SKU-level monitoring for high-revenue products to support inventory planning and promotion decisions.

10. Conclusion

The project demonstrates a complete data-analytics workflow: loading and cleaning data, exploratory analysis, visualization, insight generation and business recommendations.

Dataset Source

<https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data>